

Extra Practice: Pandas & OOP

Practice questions on group_by, pivot, indexing, and object-oriented programming — the topics flagged for review. No guarantee these match the final exam.

For all questions, you may assume:

```
import numpy as np
import pandas as pd
```

Q1. What is printed?

```
class Dog:
    def __init__(self, name):
        self.name = name
    def __str__(self):
        return f"Dog named {self.name}"
    def __repr__(self):
        return f"Dog({self.name})"
```

```
d = Dog("Rex")
print(d)
print([d])
```

- a.
Dog named Rex
[Dog named Rex]
- b.
Dog named Rex
[Dog(Rex)]
- c.
Dog(Rex)
[Dog(Rex)]
- d.
Dog named Rex
[<__main__.Dog object at 0x1f3>]

Q2. What is printed?

```
class Counter:
    def __init__(self):
        self.count = 0
    def bump(self):
        self.count += 1
```

```
c = Counter()
c.bump()
c.bump()
print(c.count)
```

- a. 0
- b. 1
- c. 2
- d. None

Q3. What is printed?

```
class Money:
    def __init__(self, dollars=0):
        self.dollars = dollars
    def __add__(self, other):
        return Money(self.dollars + other.dollars)
```

```
def __str__(self):
    return f"${self.dollars}"
```

```
print(Money(5) + Money(3))
```

- a. \$8
- b. 8
- c. Money(8)
- d. A TypeError is raised.

Q4. What is printed?

```
class Animal:
    def speak(self):
        return "..."
class Cat(Animal):
    def speak(self):
        return "meow"
class Fish(Animal):
    pass
```

```
print(Cat().speak())
print(Fish().speak())
```

- a.
meow
meow
- b.
meow
AttributeError: 'Fish' object has no attribute 'speak'
- c.
...
...
- d.
meow
...

Q5. What is printed?

```
class Bag:
    def __init__(self, items):
        self.items = items
    def __len__(self):
        return len(self.items)
    def __getitem__(self, i):
        return self.items[i]
```

```
b = Bag([10, 20, 30])
print(len(b))
print(b[1])
```

- a. 3 / 10
- b. 3 / 20
- c. A TypeError is raised.
- d. 3 / [10, 20, 30]

Q6. What is printed?

```
class Box:
    def __init__(self):
        self.value = 1
def modify(box):
```

```
box.value = 99
```

```
b = Box()
modify(b)
print(b.value)
```

- a. 1
- b. None
- c. 99
- d. A TypeError is raised.

Q7. What is printed?

```
df = pd.DataFrame({"team": ["A", "B", "A", "B"],
                  "pts": [10, 20, 30, 40]})
print(df.groupby("team")["pts"].sum())
```

- a.
team
A 40
B 60
Name: pts, dtype: int64
- b.
team
A 20
B 30
Name: pts, dtype: int64
- c. 100
- d.
team
A 10
B 20
Name: pts, dtype: int64

Q8. What is printed?

```
df = pd.DataFrame({"g": ["x", "x", "y", "x"],
                  "v": [1, 2, 3, 4]})
print(df.groupby("g").size())
```

- a.
g
x 2
y 1
dtype: int64
- b.
g
x 3
y 3
dtype: int64
- c. 4
- d.
g
x 3
y 1
dtype: int64

Q9. What is printed?

```
df = pd.DataFrame({"grp": ["a", "a", "b"],
                  "score": [4, 8, 10]})
result = df.groupby("grp")["score"].mean()
```

```
print(result.loc["a"])
a. 12.0
b. 6.0
c. 7.333333333333333
d. 10.0
```

Q10. What is printed?

```
df = pd.DataFrame({"dept": ["sales", "eng", "sales", "eng"],
                    "pay": [50, 90, 70, 110]})
print(df.groupby("dept")["pay"].max())
```

- a.
dept
sales 70
eng 110
Name: pay, dtype: int64
- b.
dept
eng 90
sales 50
Name: pay, dtype: int64
- c.
dept
eng 110
sales 70
Name: pay, dtype: int64
- d. 110

Q11. What is printed?

```
df = pd.DataFrame({"day": ["Mon", "Mon", "Tue"],
                    "item": ["a", "b", "a"],
                    "sales": [10, 20, 40]})
print(df.pivot_table(index="day", values="sales", aggfunc="sum"))
```

- a.
 sales
day
Mon 30
Tue 40
- b.
 sales
day
Mon 15
Tue 40
- c.
 sales
day
Mon 10
Tue 40
- d.
 sales
day
Mon 30
Tue 30

Q12. What is printed?

```
df = pd.DataFrame({"row": ["r1", "r1", "r2"],
                    "col": ["c1", "c2", "c1"]},
```

```

        "val": [1, 2, 3]})
print(df.pivot(index="row", columns="col", values="val"))

```

a.

```

col  c1  c2
row
r1    1   2
r2    3   0

```

b. A ValueError is raised.

c.

```

col  c1  c2
row
r1  1.0  2.0
r2  3.0  0.0

```

d.

```

col  c1  c2
row
r1  1.0  2.0
r2  3.0  NaN

```

Q13. What is printed?

```

df = pd.DataFrame({"cat": ["x", "x", "y"],
                    "n": [2, 6, 10]})
print(df.pivot_table(index="cat", values="n"))

```

a.

```

      n
cat
x      8
y     10

```

b.

```

      n
cat
x    4.0
y   10.0

```

c.

```

      n
cat
x      2
y     10

```

d. A ValueError is raised because no aggfunc was given.

Q14. What is printed?

```

df = pd.DataFrame({"a": ["x", "x", "y"],
                    "b": ["p", "q", "p"],
                    "v": [1, 2, 3]})
print(df.groupby(["a", "b"])["v"].sum().unstack())

```

a.

```

b    p    q
a
x    1    2
y    3    0

```

b. A ValueError is raised.

c.

```

b    p    q
a
x  1.0  2.0
y  3.0  NaN

```

d.

	b	p	q
a			
x	1.0	2.0	
y	3.0	0.0	

Q15. What is printed?

```
s = pd.Series([1, 2, 3], index=["a", "b", "c"])
print(s.reindex(["c", "a", "d"]))
```

a.

c	3.0
a	1.0
d	NaN

dtype: float64

b.

c	3
a	1
d	0

dtype: int64

c.

a	1
b	2
c	3

dtype: int64

d. A KeyError is raised because 'd' is missing.